

LINEAR INDEPENDENCE FOR A RESONANT FAMILY OF FOUR GABOR SHIFTS

PRINCIPIA MATH

ABSTRACT. Let $a \notin \mathbb{Q}$ and $j \in \mathbb{Z} \setminus \{-1, 1\}$. We prove that for every $0 \neq g \in L^2(\mathbb{R})$ the functions $g, T_1g, M_1g, M_{a+j}T_ag$ are linearly independent. This includes the Heil–Speegle configuration $\{(0, 0), (1, 0), (0, 1), (\sqrt{2}, \sqrt{2})\}$, with no regularity assumption on the window. Oussa’s Zak-transform fibration reduces a putative dependence to cocycles over irrational rotations. A theorem of Iwanik–Lemańczyk–Rudolph forces a quadratic root count on almost every nonzero fibre; Jensen’s formula then fixes the modulus of one root, which can occur on only finitely many fibres.

For $x, \omega \in \mathbb{R}$, write

$$(T_xg)(t) = g(t - x), \quad (M_\omega g)(t) = e^{2\pi i \omega t} g(t).$$

The original HRT conjecture has recently been disproved by a twelve-point Schwartz example [1], but its universal four-point form remains open. Guan and Okoudjou proved that form for real-valued windows [2]. We prove the following complementary result for an arbitrary complex window and a resonant non-lattice family.

Theorem 1. *Let $a \in \mathbb{R} \setminus \mathbb{Q}$ and $j \in \mathbb{Z} \setminus \{-1, 1\}$. For every nonzero $g \in L^2(\mathbb{R})$, the four functions*

$$g, \quad T_1g, \quad M_1g, \quad M_{a+j}T_ag$$

are linearly independent.

The case $(a, j) = (\sqrt{2}, 0)$ is the four-point configuration singled out by Heil and Speegle [4]. The Zak reduction and invariant-circle decomposition below are part of Oussa’s mixed-integer program [6]. The additional ingredients are the root-count interpretation, the use of Jensen’s formula across fibres, and the fixed-radius endgame.

We use the following consequence of Iwanik–Lemańczyk–Rudolph [5]: if a is irrational and $\xi : \mathbb{T} \rightarrow \mathbb{T}$ is absolutely continuous, has nonzero degree, and has derivative of bounded variation, then

$$U_{\xi, a}f(t) = \xi(t)f(t - a)$$

has no nonzero eigenfunction in $L^2(\mathbb{T})$. We apply this only to real-analytic ξ .

Proof of Theorem 1. Assume that

$$(1) \quad Ag + BT_1g + CM_1g + DM_{a+j}T_ag = 0.$$

By the three-point theorem [3], all four coefficients are nonzero. Let

$$Zg(t, \omega) = \sum_{n \in \mathbb{Z}} g(t - n) e^{2\pi i n \omega}$$

be the Zak transform, interpreted by unitary extension on L^2 . Its standard covariance relations turn (1) into

$$(2) \quad (A + Be^{-2\pi i \omega} + Ce^{2\pi i t})G(t, \omega) = -De^{2\pi i(a+j)t}G(t - a, \omega - a),$$

where $G = Zg$. The translation on the right preserves $\theta = \omega - t \pmod{1}$. Put

$$G_\theta(t) = G(t, t + \theta), \quad P_\theta(t) = A + Be^{-2\pi i(t+\theta)} + Ce^{2\pi i t}.$$

For almost every θ ,

$$(3) \quad G_\theta(t + 1) = e^{2\pi i(t+\theta)} G_\theta(t),$$

$$(4) \quad G_\theta(t - a) = -D^{-1}e^{-2\pi i(a+j)t} P_\theta(t) G_\theta(t),$$

and

$$(5) \quad \|g\|_2^2 = \int_{\mathbb{T}} \int_0^1 |G_\theta(t)|^2 dt d\theta.$$

Hence the set Θ of parameters for which $G_\theta \neq 0$ in $L^2(\mathbb{T})$ has positive measure. Since $|G_\theta|$ is periodic and P_θ has only finitely many zeros, the zero set of G_θ is invariant modulo null sets under the irrational rotation $t \mapsto t - a$. Thus, for almost every $\theta \in \Theta$, one has $G_\theta(t) \neq 0$ almost everywhere.

Write $z = e^{2\pi i t}$ and

$$(6) \quad Q_\theta(z) = Cz^2 + Az + Be^{-2\pi i \theta}, \quad P_\theta(t) = z^{-1}Q_\theta(z).$$

Let E be the set of θ for which Q_θ has a zero on $|z| = 1$, and for $\theta \notin E$ let $N(\theta)$ be the number of its zeros in the open unit disc. For later use, observe that for every $\rho > 0$ there are at most two θ for which Q_θ has a zero of modulus ρ . Indeed, such a zero lies on the two circles

$$|z| = \rho, \quad |A + Cz| = |B|/\rho,$$

whose centres are distinct because $A \neq 0$; once z is fixed, (6) determines θ . In particular, E is finite.

We now prove the essential identity

$$(7) \quad N(\theta) = j + 1$$

for almost every $\theta \in \Theta \setminus E$. Set

$$m_\theta(t) = e^{2\pi i \theta t} e^{\pi i t(t-1)}, \quad V_\theta = G_\theta / m_\theta.$$

Then V_θ is periodic, and (4) becomes

$$V_\theta(t - a) = \Lambda_\theta e^{-2\pi i j t} P_\theta(t) V_\theta(t)$$

for some $\Lambda_\theta \neq 0$. Since $V_\theta \neq 0$ almost everywhere, $W_\theta = V_\theta / |V_\theta|$ satisfies

$$W_\theta(t - a) = \xi_\theta(t) W_\theta(t), \quad \xi_\theta(t) = \frac{\Lambda_\theta}{|\Lambda_\theta|} e^{-2\pi i j t} \frac{P_\theta(t)}{|P_\theta(t)|}.$$

The map $\xi_\theta : \mathbb{T} \rightarrow \mathbb{T}$ is real analytic, and the argument principle gives

$$(8) \quad \deg \xi_\theta = -j + \text{wind}(P_\theta, 0) = -j - 1 + N(\theta).$$

If this degree were nonzero, the theorem quoted above would imply that $U_{\xi_\theta, a}$ has no eigenfunction. But

$$U_{\xi_\theta, a} \overline{W_\theta} = \overline{W_\theta},$$

contradicting that theorem. Thus (8) vanishes, proving (7).

If $|j| \geq 2$, then (7) is impossible because $N(\theta) \in \{0, 1, 2\}$. Consequently Θ is contained, up to a null set, in the finite set E , contradicting (5).

It remains to take $j = 0$. On almost every $\theta \in \Theta \setminus E$, the polynomial Q_θ has one zero inside and one zero $\zeta_{\text{out}}(\theta)$ outside the unit circle. Taking logarithms in (4) gives

$$\log |G_\theta(t - a)| - \log |G_\theta(t)| = \log |P_\theta(t)| - \log |D|.$$

The right side has integral zero. Indeed, whenever an L^1 function has the form $u(t - a) - u(t)$ with u finite almost everywhere, its Birkhoff sums converge to its integral almost everywhere and to zero in measure. Hence

$$(9) \quad \int_0^1 \log |P_\theta(t)| dt = \log |D|.$$

Jensen's formula and (6) now yield

$$\log |D| = \log |C| + \log |\zeta_{\text{out}}(\theta)|, \quad |\zeta_{\text{out}}(\theta)| = |D|/|C|.$$

By the fixed-radius observation above, this can occur for at most two values of θ . Thus Θ is again finite up to a null set, contradicting (5). This proves the theorem. $\square$

Remark 2. *The argument stops at $j = \pm 1$: the root count then allows both zeros to lie on the same side of the unit circle, and Jensen's formula no longer fixes the modulus of an individual root.*

REFERENCES

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